STRATEGIC RADAR

Most intelligence tools tell you what happened.
Strategic Radar tells you what it means and what to do about

Strategic Radar Deployment for Lusitex Technical Fabrics

AI-Driven Strategic Intelligence Platform for European Market Expansion

AI Infrastructure Evaluation and Implementation Strategy

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March 2026

Executive Summary

AI-Driven Strategic Intelligence Platform for European Market Expansion

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Lusitex Technical Fabrics is a Portuguese textile manufacturer seeking to expand its presence in European markets. Operating across multiple countries creates strategic challenges, including stronger competition, fragmented market intelligence, and the need to continuously monitor technological developments and business opportunities.

To support this expansion, this report proposes the deployment of Strategic Radar, an AI-powered intelligence platform designed to monitor market signals and generate structured strategic insights. The platform analyzes external data sources to detect competitor activity, identify opportunities such as public tenders and private sector projects, and track emerging textile technologies. These signals are evaluated through a retrieval-augmented generation (RAG) architecture to produce concise intelligence briefs for decision-makers.

Two infrastructure approaches were evaluated: a modular open-source architecture and a vendor-managed cloud solution based on Microsoft Azure. The comparison considered technical capabilities, operational complexity, organizational impact, risks, and cost structure.

The cost analysis shows that the open-source architecture results in a lower three-year Total Cost of Ownership (TCO) of approximately €48,240, compared with €57,160 for Azure. While Azure offers advantages in governance, security, and infrastructure management, the open-source option provides greater flexibility and lower costs, making it suitable for initial deployment.

Based on these findings, the report recommends deploying Strategic Radar using the open-source stack as a Minimum Viable Product (MVP) over eight weeks. This approach allows Lusitex to quickly establish a strategic intelligence capability while keeping flexibility to evolve the system as new requirements emerge.

By implementing Strategic Radar, Lusitex can gain continuous visibility into competitors, market opportunities, and technology trends across Europe.

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1. Client Context

Lusitex aims to expand into European markets but lacks a structured capability to continuously monitor competitors, technologies, and business opportunities.

1.1 Lusitex Business Context

Lusitex Technical Fabrics is a mid-sized Portuguese manufacturer specializing in high-performance textile materials used in sectors such as hospitality, retail environments, and defense and public security, including garments and technical fabrics for police and military organizations.

The company has developed strong capabilities in producing durable and specialized textile solutions designed for demanding commercial and institutional environments. Over time, Lusitex has built a solid position in the Portuguese market and established relationships with clients across several sectors.

However, growth opportunities within the domestic market have slowed in recent years as the Portuguese market has become increasingly saturated. As a result, Lusitex has begun pursuing a strategy of international expansion, with a particular focus on European markets.

Successfully expanding across Europe requires a deeper understanding of new competitors, emerging technologies, and potential business opportunities across multiple countries and industries.

1.2 Strategic Intelligence Challenge

Expanding into European markets significantly increases the complexity of Lusitex's competitive environment. Instead of operating primarily within the Portuguese market, the company must now monitor competitors, customers, and industry developments across multiple countries.

One key challenge is tracking competitor activity across Europe. Competitors may introduce new products, form strategic partnerships, expand geographically, or pursue acquisitions that could influence Lusitex's market position.

Another important challenge is identifying new business opportunities. These include public procurement tenders related to textiles and materials published in European tender databases, as well as private-sector opportunities such as company expansions, new commercial developments, or sector growth that may require textile solutions.

In addition, Lusitex must remain aware of emerging textile technologies, materials innovation, and relevant industry events such as textile fairs and trade shows. These developments may influence product development, partnership opportunities, and long-term competitive positioning.

To address these challenges, Lusitex plans to implement the Strategic Radar intelligence system, which will continuously monitor external information sources and transform relevant signals into structured strategic insights to support decision-making during the company's European expansion.

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2. Strategic Radar Platform

Strategic Radar is an AI-powered intelligence platform designed to monitor market signals and generate structured strategic insights.

2.1 Overview

Strategic Radar is an AI-powered strategic intelligence platform designed to help organizations continuously monitor external market developments and transform them into actionable insights. The system automatically collects signals from multiple external information sources, including news, industry websites, and public data platforms.

These signals may relate to competitor activity, business opportunities, or emerging technologies. Once collected, they are evaluated using a Retrieval-Augmented Generation (RAG) architecture that combines large language models with a structured knowledge base containing information about the company's strategy, products, competitors, and technology priorities.

Based on this evaluation, the system generates concise intelligence briefs that summarize the signal, assess its potential strategic relevance, and suggest possible actions. By automating the monitoring and analysis of external signals, Strategic Radar allows decision-makers to detect relevant developments earlier and respond more quickly to changes in the market environment.

2.2 Main Features and Functionality

2.2.1 Competitor Intelligence Monitoring

The system continuously monitors competitor organizations and identifies strategic signals that may affect the company's competitive position. These signals may include partnerships, acquisitions, product launches, leadership changes, investments, or geographic expansion.

Information is collected through web search, news monitoring services, and other online sources. The system then evaluates the relevance of these signals and highlights developments that may require strategic attention.

2.2.2 Business Opportunity Detection

Strategic Radar detects potential commercial opportunities by monitoring multiple external sources related to both public and private sector developments.

This includes public procurement tenders published in European databases, as well as signals indicating new private-sector projects, company expansions, or sector growth that may require textile solutions. The system evaluates whether these opportunities align with Lusitex's products, capabilities, and target markets.

2.2.3 Technology Trend Monitoring

The platform also monitors technological developments relevant to the textile and materials sector. Signals related to new materials, manufacturing techniques, or emerging textile technologies are detected and evaluated.

These signals are compared against a predefined technology watchlist to identify potential innovation opportunities, emerging industry trends, or technological risks that may affect the company's future competitiveness.

2.2.4 RAG-Based Strategic Evaluation

All detected signals are evaluated using a Retrieval-Augmented Generation (RAG) architecture. This approach combines large language models with a structured knowledge base containing information about the company's strategic priorities, product portfolio, competitor landscape, and technology interests.

By grounding the analysis in this contextual knowledge, the system produces more accurate and strategically relevant intelligence outputs.

2.2.5 Executive Intelligence Brief Generation

For signals considered strategically relevant, the system generates concise executive intelligence briefs. These summaries highlight the key development, explain its potential impact on the company, and may include suggested actions or areas for further investigation.

The goal is to provide decision-makers with clear and quickly digestible insights that support informed strategic decisions.

2.2.6 Automated Intelligence Delivery

The platform integrates with workflow automation tools to distribute intelligence outputs automatically. Intelligence briefs can be delivered through scheduled reports or sent directly via email notifications.

This automated delivery ensures that relevant information reaches decision-makers in a timely manner without requiring manual monitoring of multiple information sources.

3. Strategic Radar System Architecture

The platform uses a modular AI architecture combining APIs, vector search, and automated workflows to process market signals and generate intelligence briefs.

3.1 Architecture Overview

The Strategic Radar platform is implemented as a modular AI system composed of independent infrastructure components that work together through an API-driven architecture. Each layer of the system is responsible for a specific function, such as signal collection, intelligence analysis, data storage, or automated delivery.

This modular design allows the platform to remain flexible and extensible. Individual components can be replaced or upgraded without requiring a full redesign of the system, which is important for AI-based systems that may evolve as new technologies and tools become available.

At the core of the platform is a Retrieval-Augmented Generation (RAG) pipeline. External signals are first collected from various data sources and then evaluated using a large language model that retrieves contextual information from a structured knowledge base. This approach ensures that the AI analysis is grounded in the strategic context of the organization, including company strategy, products, competitors, and technology priorities.

Together, these components form an integrated intelligence pipeline that continuously collects signals, evaluates their relevance, and produces structured intelligence outputs for decision-makers.

3.2 Infrastructure Components and Technology Selection

The infrastructure architecture of Strategic Radar follows a modular open-stack approach in which each layer of the system is implemented using specialized technologies. The selection of tools was primarily influenced by previous experience with the technologies and the objective of minimizing implementation risk.

By relying on tools already familiar to the development team, the project avoided unnecessary complexity related to learning new infrastructure while building a complex AI system. This allowed the development effort to focus primarily on building the intelligence functionality of the platform rather than solving infrastructure-related challenges.

3.2.1 FastAPI — Backend API Layer

What it does

FastAPI provides the backend API layer of the Strategic Radar system. It exposes the endpoints that trigger the intelligence pipeline, receive requests from the user interface, and return the generated intelligence briefs.

Why it was chosen

FastAPI was selected primarily because of its seamless integration with Python, which is the main programming language used throughout the system. Since the AI pipeline, agent orchestration, and data processing components are implemented in Python, FastAPI allows the backend layer to integrate directly with the rest of the system without requiring additional interoperability layers.

3.2.2 LangGraph — Agent Workflow Orchestration

What it does

LangGraph orchestrates the internal workflow of the Strategic Radar intelligence pipeline. It defines the sequence of steps required to generate an intelligence report, including signal collection, signal evaluation, and the generation of the final executive brief.

The workflow is implemented as a structured pipeline where each node performs a specific task and passes the results to the next stage of the process.

Why it was chosen

LangGraph was selected because it enables the creation of structured and extensible AI workflows. Although the current MVP implementation includes relatively simple workflows, the system was designed to support future expansion of the intelligence pipeline.

Another important advantage is the ability to maintain deterministic workflows. Compared to fully autonomous agent systems, LangGraph allows tighter control

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over execution order, improving reliability and predictability—an important characteristic for decision-support systems.

3.2.3 Pinecone — Vector Database (RAG System)

What it does

Pinecone serves as the vector database used in the Retrieval-Augmented Generation (RAG) architecture. It stores vector embeddings representing the knowledge base used by the system, including the company profile, strategic objectives, competitor registry, and technology watchlist.

During signal evaluation, the system retrieves relevant context from this knowledge base and provides it to the language model to ground the analysis.

Why it was chosen

Pinecone was selected because it is a fully managed vector database service that simplifies deployment and operation. Using a managed service avoids the need to deploy and maintain a database infrastructure manually.

Previous experience with the platform also helped reduce implementation risk and enabled faster development of the RAG architecture.

3.2.4 OpenAI — Large Language Model Provider

What it does

OpenAI provides the large language models used by Strategic Radar to analyze collected signals and generate executive intelligence briefs. The models evaluate the strategic relevance of detected signals and produce structured summaries designed for decision-makers.

Why it was chosen

The OpenAI API was selected primarily because API access was available within the Ironhack learning environment. This allowed the project to integrate large language models without requiring additional infrastructure or model configuration.

The API integrates easily with Python applications and provides the reasoning and text-generation capabilities required by the intelligence evaluation pipeline.

3.2.5 n8n — Workflow Automation

What it does

n8n is used to automate operational workflows related to intelligence delivery. It manages the final stages of the pipeline, including receiving the generated intelligence brief, formatting it as an HTML email, and sending the report to users.

The system also uses n8n to schedule automated intelligence runs, triggering periodic analysis of monitored competitors, technologies, and opportunity sources.

Why it was chosen

The use of n8n was partly influenced by project requirements within the Ironhack program, which included workflow automation as part of the system architecture. Within the project, n8n demonstrates the system's ability to integrate automated workflows and operational orchestration without requiring additional backend development.

3.2.6 Railway — Application Hosting

What it does

Railway provides the hosting environment for the Strategic Radar backend application. It runs the FastAPI server and the AI workflow pipeline, allowing the system to operate continuously and expose the API endpoints used by the automation workflows.

Why it was chosen

Railway was selected because it simplifies the deployment and management of backend applications. It allows the system to be deployed quickly without requiring manual infrastructure configuration.

Another advantage is its integration with GitHub, which enables automatic deployment whenever new code is pushed to the repository. This significantly simplifies the development workflow and supports rapid iteration during development.

3.2.7 SendGrid — Email Delivery

What it does

SendGrid is used to deliver the intelligence reports generated by the Strategic Radar system via email. After the intelligence brief is created, the report is formatted and distributed automatically to users.

The delivery process is orchestrated by n8n, which handles the formatting and sending of the messages through the SendGrid infrastructure.

Why it was chosen

SendGrid was selected because it integrates easily with n8n and provides reliable email delivery capabilities. The availability of a free tier also made it suitable for development and demonstration purposes without introducing additional infrastructure costs.

3.2.8 External Data Sources — Tavily, NewsAPI, HackerNews, and TED

What they do

Strategic Radar collects signals from several external data sources in order to identify developments that may affect the company's competitive environment.

The system currently integrates four main sources:

- **Tavily API**, used to perform web searches and discover information about competitors, technologies, and market developments.
- **NewsAPI**, which provides structured access to news articles from multiple publications, enabling detection of industry announcements and developments.
- **HackerNews**, used to capture early signals related to new technologies, tools, and innovations discussed within the technology community.
- **TED (Tenders Electronic Daily)**, the official European Union public procurement database, used to identify public-sector tenders that may represent potential business opportunities.

Why they were chosen

The selection of these sources was primarily driven by risk management considerations and prior experience with the tools. Using familiar APIs reduced integration time and minimized development risk during the project.

At the same time, the inclusion of TED allowed the system to incorporate public procurement signals relevant to business development, while HackerNews provided technology-focused signals that complement general web and news sources.

Together, these sources enable the system to capture multiple categories of strategic signals, including competitor activity, industry developments, emerging technologies, and public procurement opportunities.

4. Vendor Cloud Architecture Alternative

Two infrastructure approaches were evaluated: a modular open-source stack and a vendor-managed cloud platform based on Microsoft Azure.

4.1 Vendor Selection

Before selecting a vendor cloud platform for comparison with the open-source infrastructure stack, several major cloud providers were evaluated. The analysis focused on platforms capable of supporting enterprise-grade AI applications, including infrastructure for hosting backend services, running AI workloads, and integrating large language models.

The main cloud platforms considered were:

- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)
- Microsoft Azure

These three providers dominate the global cloud infrastructure market and offer similar technical capabilities for building AI-powered systems. Each platform provides services for application hosting, machine learning deployment, workflow automation, vector search, and scalable data processing.

Although all three platforms are technically capable of supporting the Strategic Radar architecture, **Microsoft Azure** was selected as the reference vendor platform for the infrastructure comparison. The decision was based primarily on its strong enterprise adoption in Portugal and across Europe, as well as its tight integration with the Microsoft ecosystem commonly used by many organizations.

4.2 Microsoft Azure

For the infrastructure comparison in this project, Microsoft Azure was selected as the representative vendor-managed cloud platform to compare against the open-source infrastructure stack used in the Strategic Radar system.

The selection of Azure reflects several strategic and practical considerations relevant to organizations operating in the Portuguese and European enterprise environments.

4.2.1 Enterprise Adoption

Microsoft Azure has a strong presence among Portuguese enterprises and public institutions. Many organizations already rely on Microsoft technologies such as Microsoft 365, Windows Server, and Active Directory for their core IT infrastructure.

Because of this existing footprint, Azure is often a natural extension of the corporate technology stack and is frequently chosen for digital transformation initiatives, particularly when security, governance, and enterprise integration are important.

4.2.2 Microsoft Ecosystem Integration

Azure integrates directly with widely used Microsoft enterprise tools, including:

- Microsoft 365
- Azure Active Directory
- Power BI
- Microsoft Defender security services

This integration allows organizations to manage identity, access control, monitoring, and security policies within a unified ecosystem. For companies already operating within the Microsoft environment, Azure can significantly reduce integration complexity when deploying new applications or AI solutions.

4.2.3 Azure OpenAI

Azure provides access to OpenAI models through the **Azure OpenAI Service**, enabling organizations to use advanced language models such as GPT-based systems within a managed enterprise environment.

This service provides access to the same model capabilities available through the OpenAI API while adding enterprise features such as:

- private networking
- enterprise security controls
- compliance certifications
- centralized billing and monitoring

Since the Strategic Radar system relies heavily on large language models for signal evaluation and intelligence generation, Azure OpenAI represents the natural vendor-managed equivalent of the OpenAI API used in the open-source architecture.

4.2.4 Compliance and Security

Azure provides a wide range of enterprise compliance certifications that are particularly relevant for European organizations, including:

- ISO 27001
- SOC 2
- GDPR compliance support

These certifications are important for systems that process strategic intelligence data or potentially sensitive business information. Using a managed cloud platform with established compliance frameworks can simplify governance and risk management.

4.2.5 Managed Services

Another advantage of Azure is the availability of managed infrastructure services that reduce operational complexity. Examples include:

- Azure App Service for application hosting
- Azure Logic Apps for workflow automation
- Azure AI Search for vector search and knowledge retrieval
- Azure Monitor for system monitoring and logging

Because the client context specifies that the organization does not have a dedicated DevOps team, these managed services can significantly simplify system maintenance and operational management. The platform allows infrastructure to be managed through graphical dashboards and vendor tools rather than requiring extensive infrastructure engineering expertise.

4.3 Infrastructure Comparison

The following table compares the components used in the current open-source infrastructure stack with the equivalent services available in the Azure ecosystem.

Current Component	Function in the System	Azure Equivalent	Key Differences	Notes
FastAPI	Backend API exposing endpoints that trigger the intelligence pipeline and return results	Azure App Service	Managed hosting for web applications with built-in scaling and monitoring	Same Python application could be deployed
LangGraph	Orchestrates the AI workflow pipeline and controls the sequence of intelligence processing steps	Azure Functions / Azure Container Apps	Serverless or container-based execution	Workflow logic would remain in application code
Pinecone	Vector database storing embeddings used for RAG signal evaluation	Azure AI Search (Vector Search)	Vector search integrated into Azure managed search service	Supports embedding storage and retrieval
OpenAI API	Provides LLM models used for signal evaluation and intelligence brief generation	Azure OpenAI Service	Same models hosted within Azure infrastructure	Adds enterprise security and compliance
n8n	Workflow automation for scheduling runs and orchestrating report delivery	Azure Logic Apps	Fully managed workflow automation integrated with Azure services	Similar automation capabilities

Current Component	Function in the System	Azure Equivalent	Key Differences	Notes
Railway	Hosting environment for backend application and AI pipeline	Azure App Service / Azure Container Apps	Enterprise-grade hosting with autoscaling and SLA guarantees	Azure replaces third-party hosting
SendGrid	Email delivery of intelligence reports	SendGrid (Azure partner service) / Azure Communication Services	Integrated within Azure ecosystem	SendGrid often used alongside Azure
Pinecone + Markdown KB	Knowledge base used for RAG grounding	Azure AI Search + Azure Blob Storage	Managed storage combined with vector search	Enterprise storage layer
Tavily / NewsAPI	External market signal discovery	Same APIs	External sources remain unchanged	Azure only hosts the application
TED (EU tenders)	Public procurement signals used for opportunity detection	Same data source	External system	No Azure equivalent required

5. External Environment Analysis

The external environment affects infrastructure selection through regulatory, technological, and economic factors.

This chapter analyzes the external factors that may influence the adoption and operation of the Strategic Radar platform using the PESTLE framework. The analysis considers six dimensions—Political, Economic, Social, Technological, Legal, and Ecological—to evaluate the broader environment surrounding the infrastructure choices.

Because the project considers two alternative infrastructure approaches—a modular open-stack architecture and a vendor-managed cloud platform—each option is analyzed separately. This allows the comparison to highlight how external factors may affect the feasibility, risks, and long-term sustainability of each approach.

5.1 PESTLE — Open Stack

This section examines how external environmental factors influence the adoption of the modular open-stack architecture used in the Strategic Radar system. The analysis considers how political, economic, technological, and regulatory conditions may affect the use of distributed services, open-source tools, and multiple external providers.

The open-stack approach emphasizes flexibility, modularity, and the use of specialized tools within the AI ecosystem. While this architecture enables rapid innovation and experimentation, it also introduces considerations related to governance, operational complexity, and regulatory compliance.

Factor	Open-Source Stack Considerations	Implications for the Project
Political	The system relies on globally distributed cloud services such as Railway, Pinecone, and OpenAI, as well as external APIs. Data sovereignty concerns may arise if data is processed or stored outside the European Union. At the same time, the European technology landscape increasingly promotes open digital ecosystems and technological independence.	Organizations operating in regulated sectors may require guarantees about EU data residency. However, open architectures can also align with European initiatives promoting technological autonomy and reduced dependence on large cloud vendors.
Economic	Modular open-stack architectures typically reduce initial infrastructure costs because tools can be selected individually and many services offer free or low-cost tiers. However, long-term operational costs may become less predictable as usage grows across multiple services.	The approach is well suited for experimentation, innovation, and early-stage deployments. As the system scales, cost management may require careful monitoring across multiple service providers.
Social	The open-source ecosystem benefits from a large global developer community and strong innovation culture. Many AI practitioners prefer modular architectures that allow experimentation with different tools and models. However, enterprise organizations may sometimes prefer vendor-supported platforms with formal support structures.	The architecture may be particularly attractive for technically oriented teams and innovation-driven environments, while some enterprises may require additional governance or support arrangements.
Technological	Open architectures allow the integration of specialized tools such as Pinecone, LangGraph, and n8n. This provides high flexibility and enables rapid adoption of new AI technologies. However, integrating multiple independent tools increases system	The architecture offers strong innovation potential and adaptability but requires greater technical expertise for system integration, maintenance, and evolution.

Factor	Open-Source Stack Considerations	Implications for the Project
	complexity and may require more engineering effort.	
Legal	European regulations such as GDPR impose strict requirements on how data is processed and stored. Using multiple third-party services may introduce additional complexity in ensuring compliance and managing data processing agreements.	Organizations must implement governance policies and data management practices to ensure regulatory compliance across all external services used by the system.
Ecological	Distributed infrastructure across several providers may make it more difficult to evaluate the environmental impact of the system. Smaller service providers may offer limited sustainability reporting compared to large cloud vendors.	Sustainability analysis may require additional effort to assess the environmental footprint of each infrastructure component.

5.2 PESTLE — Azure

This section analyzes the same external factors from the perspective of a vendor-managed cloud platform, using Microsoft Azure as the reference environment. Vendor cloud platforms provide integrated infrastructure services designed to simplify deployment, security management, and operational governance.

The analysis evaluates how Azure's enterprise ecosystem, managed services, and compliance frameworks influence the feasibility of deploying the Strategic Radar system within a centralized cloud environment commonly used by large organizations.

Factor	Azure Vendor Solution Considerations	Implications for the Project
Political	Microsoft operates cloud regions across Europe and supports EU data residency requirements. Azure infrastructure is widely adopted by European enterprises and public institutions.	Using Azure may simplify alignment with European data governance expectations and reduce concerns related to cross-border data transfers.
Economic	Vendor platforms typically involve higher infrastructure costs compared to modular open solutions, particularly when using managed AI services. However, centralized billing and integrated services may simplify financial management.	Organizations must balance higher infrastructure costs against the operational efficiency and predictability provided by managed services.
Social	Azure has strong adoption within enterprise environments, particularly in organizations already using Microsoft technologies such as Microsoft 365 and Azure Active Directory.	Adoption may be smoother in organizations that already rely on Microsoft infrastructure and established enterprise IT standards.
Technological	Azure provides integrated managed services such as Azure OpenAI, Azure AI Search, and Azure Logic Apps. These services reduce the need for custom infrastructure integration but may reduce architectural flexibility.	The platform simplifies operations and system maintenance but may introduce a degree of vendor dependency.
Legal	Azure provides numerous compliance certifications including ISO 27001, SOC 2, and GDPR compliance frameworks. Microsoft also offers contractual and legal documentation commonly required by enterprise procurement processes.	Compliance and regulatory governance may be easier to manage for organizations operating in regulated environments.

Factor	Azure Vendor Solution Considerations	Implications for the Project
Ecological	Microsoft publishes sustainability commitments and environmental reporting related to its data centers and energy usage. Many Azure regions operate with renewable energy targets.	Organizations that prioritize sustainability reporting may benefit from the transparency provided by large cloud providers.

5.3 Key Insights

The PESTLE analysis highlights that both infrastructure approaches present distinct advantages and trade-offs depending on organizational priorities, technical capabilities, and governance requirements.

From a **political and regulatory perspective**, vendor-managed platforms such as Azure provide clearer guarantees regarding data residency, compliance certifications, and enterprise governance. However, open-stack architectures can also align with broader European goals of technological independence and reduced reliance on large technology providers.

From an **economic perspective**, the open-stack architecture generally offers lower initial infrastructure costs and greater flexibility in selecting services. Vendor platforms may involve higher operational costs but provide more centralized billing structures and simplified cost management.

From a **social and organizational standpoint**, open architectures tend to appeal to technically oriented teams and innovation-driven environments that value flexibility and experimentation. In contrast, vendor platforms such as Azure may align better with organizations that already operate within established enterprise IT ecosystems.

From a **technological perspective**, open architectures offer greater flexibility and the ability to integrate specialized tools across the rapidly evolving AI ecosystem. Vendor platforms emphasize integration, managed services, and operational simplicity, which can reduce infrastructure management requirements.

From a **legal and compliance standpoint**, vendor platforms typically provide more structured frameworks for meeting regulatory requirements, while open architectures require additional governance to ensure compliance across multiple service providers.

Finally, from an **ecological perspective**, large cloud providers often provide clearer sustainability reporting and environmental commitments. However, open architectures allow organizations to selectively adopt infrastructure providers aligned with their own sustainability priorities.

Overall, the choice between the two approaches depends on the organization's priorities: **flexibility and cost efficiency with the open-stack architecture, or operational simplicity and enterprise governance with the vendor-managed platform.**

6. Organizational Impact at Lusitex

Deploying Strategic Radar introduces organizational changes affecting stakeholders across sales, business development, IT, and management.

The implementation of the Strategic Radar platform affects multiple areas of the Lusitex organization. Because the system introduces AI-driven intelligence capabilities that support strategic decision-making, its impact extends beyond the IT department to commercial, operational, and innovation functions.

This chapter analyzes the key stakeholders involved in the project, their level of influence and interest, and the concerns they may have regarding the adoption of the system. Understanding these stakeholder dynamics is essential to ensure successful implementation, organizational alignment, and long-term adoption of the platform.

6.1 Stakeholder Analysis

The Strategic Radar system involves several stakeholders across Lusitex's organizational structure. These stakeholders can be grouped into four main functional areas: strategic and commercial leadership, operational and technical teams, financial governance, and innovation functions.

Each stakeholder has a different role in the project and may be affected by the system in different ways.

6..1.1 Stakeholder Identification

Stakeholder	Role in the Project
Founder / CEO	Defines company strategy and approves the investment in the Strategic Radar system to support international expansion.
Business Development Director	Responsible for identifying and developing new business opportunities in European markets using intelligence insights.
Sales Manager & Sales Team	Use intelligence outputs to identify potential clients, partnerships, and procurement opportunities.
Marketing Manager	Uses intelligence signals to identify relevant textile fairs, industry events, and branding opportunities across Europe.
IT Manager & Internal Tools Team	Responsible for deploying, integrating, and maintaining the Strategic Radar platform within the company's internal systems.
Finance Manager	Evaluates infrastructure costs, manages budgets, and assesses the financial return on investment of the system.
Production Manager	Ensures manufacturing capacity can respond to new market opportunities identified by the intelligence platform.
Procurement Manager	Monitors supplier developments and evaluates supply chain implications associated with new opportunities.
Product Development & Design Team	Uses technology intelligence signals to identify new textile materials, innovations, and product development opportunities.

6.1.2 Stakeholder Influence, Interest, and Concerns

Different stakeholders have varying levels of influence over the project and different levels of interest in the outcomes of the Strategic Radar system.

Stakeholder	Interest	Influence	Key Concerns
Founder / CEO	High	High	Strategic growth in European markets, return on investment, and positioning Lusitex as an innovative company adopting AI technologies.
Business Development Director	High	Medium	Ability to identify new opportunities across Europe and ensure AI insights remain reliable and actionable.
Sales Manager & Sales Team	High	Medium	Access to new leads and tenders; potential concern that AI could replace some manual intelligence activities currently performed by the team.
Marketing Manager	High	Medium	Identification of relevant textile fairs and industry events across Europe and using intelligence insights to support marketing strategy.
IT Manager & Internal Tools Team	High	High	System reliability, integration with internal tools, and potential changes to the role of internal IT teams due to increasing AI adoption.
Finance Manager	Medium	High	Infrastructure costs, budget predictability, and financial justification for the investment.

Stakeholder	Interest	Influence	Key Concerns
Production Manager	Medium	Medium	Ability to scale production if new market opportunities are identified by the system.
Procurement Manager	Medium	Medium	Supplier capacity and raw material availability if expansion accelerates.
Product Development & Design Team	High	Medium	Monitoring emerging textile technologies and using insights to guide product innovation.

6.1.3 Stakeholder Influence–Interest Mapping

The stakeholders can also be categorized using a **power–interest matrix**, which helps determine the level of engagement required during the project.

Stakeholder	Influence	Interest	Quadrant	Description
Founder / CEO	High	High	Quadrant I	Key decision maker responsible for approving the project and ensuring alignment with company strategy.
IT Manager & Internal Tools Team	High	High	Quadrant I	Responsible for implementing and maintaining the Strategic Radar infrastructure.
Business Development Director	Medium	High	Quadrant II	Uses intelligence insights to identify partnerships and new opportunities in European markets.
Sales Manager & Sales Team	Medium	High	Quadrant II	Use intelligence insights to identify leads and procurement opportunities.

Stakeholder	Influence	Interest	Quadrant	Description
Marketing Manager	Medium	High	Quadrant II	Uses insights to identify marketing events and branding opportunities.
Product Development & Design Team	Medium	High	Quadrant II	Monitors emerging textile technologies to support innovation.
Production Manager	Medium	Medium	Quadrant II	Ensures manufacturing capacity can support demand from new opportunities.
Finance Manager	High	Medium	Quadrant III	Oversees project budgets and financial return on investment.
Procurement Manager	Low	Low	Quadrant IV	Limited direct involvement but affected by supply chain implications of expansion.

6.2 Stakeholder Engagement Strategy

The stakeholder engagement strategy defines how the project team should communicate and collaborate with each stakeholder group during the implementation of the Strategic Radar system. Engagement intensity is determined by the stakeholder's level of influence and interest.

Quadrant	Stakeholder Type	Engagement Strategy
Quadrant I	High Influence / High Interest	Actively involve these stakeholders in project governance and decision-making. Provide regular progress updates and include them in milestone reviews and key architecture decisions.

Quadrant	Stakeholder Type	Engagement Strategy
Quadrant II	High Interest / Lower Influence	Engage frequently through demonstrations, feedback sessions, and user testing. Ensure their operational needs are reflected in system functionality and intelligence outputs.
Quadrant III	High Influence / Lower Interest	Keep these stakeholders informed through periodic reports focusing on costs, risks, and compliance aspects of the project.
Quadrant IV	Low Influence / Low Interest	Provide occasional updates and monitor for changes in influence or interest as the project evolves.

This engagement approach helps ensure that the stakeholders most critical to the success of the Strategic Radar system remain closely involved while maintaining efficient communication across the organization.

7. Strategic Radar Deployment Plan

The system will be deployed as an eight-week MVP followed by continuous monitoring and incremental improvements.

This chapter outlines how the Strategic Radar system would be implemented within Lusitex. It defines the project objectives for each infrastructure option and evaluates the main risks associated with deploying and operating the system.

Because the project evaluates two infrastructure approaches—an open modular architecture and a vendor-managed cloud solution—objectives and risks are analyzed separately for each scenario. This allows decision-makers to better understand the trade-offs associated with each implementation strategy.

7.1 Project Objectives

The project objectives define the expected outcomes of implementing the Strategic Radar system at Lusitex Technical Fabrics. These objectives follow the **Triple Constraint model**, which evaluates projects according to three key dimensions:

- **Quality** – the capabilities and reliability of the system delivered
- **Time** – the implementation timeline required to deliver the system
- **Cost** – the infrastructure and operational costs associated with deployment

Because the project evaluates two infrastructure alternatives, objectives are defined separately for each option.

7.1.1 Open Stack Objectives

Constraint	Objective
Quality	Deploy a functional Strategic Radar system capable of monitoring competitor activity, identifying business opportunities, and detecting emerging textile technologies across European markets. The system should generate reliable intelligence summaries and integrate with Lusitex internal workflows.
Time	Deliver a **Minimum Viable Product (MVP)** within **8 weeks**, including infrastructure setup, integration of data sources, implementation of the RAG pipeline, and automated generation of intelligence reports.
Cost	Maintain relatively low infrastructure costs by using modular services, open-source tools, and free-tier cloud services where possible while ensuring the system remains scalable.

7.1.2 Azure Objectives

Constraint	Objective
Quality	Deploy a robust and scalable Strategic Radar system using Azure managed services, ensuring high reliability, enterprise-grade security, and simplified infrastructure management for the operations team.
Time	Deliver a production-ready MVP deployment within **8 weeks**, leveraging managed services such as **Azure OpenAI**, **Azure AI Search**, and **Azure Logic Apps** to accelerate implementation.

Constraint	Objective
Cost	Maintain predictable infrastructure costs through centralized vendor billing while ensuring long-term scalability as the system expands to additional markets and intelligence sources.

7.2 Risk Assessment

The deployment of the Strategic Radar system introduces several technical, operational, and organizational risks. These risks vary depending on the infrastructure architecture used.

The following sections analyze the potential risks associated with each infrastructure option and propose mitigation strategies to reduce their impact.

7.2.1 Open Stack Risks

The open-stack architecture relies on a modular combination of services such as FastAPI, LangGraph, Pinecone, Railway, and multiple external APIs. While this approach provides flexibility and lower initial costs, it introduces risks related to integration complexity, operational management, and external dependencies.

Risk Category	Risk Description	Probability	Impact	Mitigation Strategy
Technical Risk	Integration challenges between multiple independent services and APIs.	Medium	Medium	Use well-documented APIs, modular architecture, and incremental testing during development.

Risk Category	Risk Description	Probability	Impact	Mitigation Strategy
Operational Risk	Lack of a dedicated DevOps team may make maintenance of multiple services more complex.	Medium	High	Prefer managed services where possible and provide clear operational documentation.
External Dependency Risk	The system depends on third-party APIs (OpenAI, Tavily, NewsAPI, TED) that may change pricing or availability.	Medium	Medium	Monitor API usage and periodically evaluate alternative providers.
AI Reliability Risk	AI-generated intelligence summaries may contain inaccuracies or hallucinations.	Medium	High	Use RAG grounding with a knowledge base and include human validation of critical insights.
Cost Management Risk	Infrastructure costs may increase unpredictably if API usage grows significantly.	Medium	Medium	Implement monitoring of API consumption and usage limits.
Organizational Risk	Employees may resist the adoption of AI tools due to concerns about automation.	Low	Medium	Communicate the system as a decision-support tool that complements existing roles.
Data Governance Risk	Use of multiple external services may raise concerns regarding data governance and compliance.	Low	Medium	Avoid transmitting sensitive data and review provider policies.

Risk Category	Risk Description	Probability	Impact	Mitigation Strategy
Scope Creep Risk	Stakeholders may request additional features once the system begins delivering intelligence reports.	Medium	Medium	Clearly define the MVP scope and prioritize additional features for future phases.
System Learning Curve Risk	The system may require time to identify the most relevant signals and produce useful insights.	High	Medium	Implement user feedback loops and allow a calibration period after deployment.

7.2.2 Azure Risks

The Azure-based infrastructure approach uses managed services such as Azure OpenAI, Azure AI Search, Azure Logic Apps, and Azure App Service. This architecture simplifies infrastructure management but introduces risks related to vendor dependency, cost structure, and platform constraints.

Risk Category	Risk Description	Probability	Impact	Mitigation Strategy
Vendor Lock-in Risk	Dependence on Azure services may make migration to other platforms difficult in the future.	Medium	High	Maintain modular architecture and abstraction layers where possible.

Risk Category	Risk Description	Probability	Impact	Mitigation Strategy
Cost Escalation Risk	Managed services may increase operational costs as system usage grows.	Medium	High	Implement cost monitoring tools and budget alerts.
Service Dependency Risk	Dependence on a single cloud provider means outages or platform changes could affect system availability.	Low	Medium	Use monitoring tools and maintain backup strategies.
Platform Constraint Risk	Some Azure services may limit customization compared to open architectures.	Low	Medium	Avoid excessive reliance on proprietary features.
AI Reliability Risk	AI-generated outputs may still contain inaccuracies or hallucinations.	Medium	High	Use RAG-based grounding and human validation of critical intelligence insights.
Organization al Adoption Risk	Employees may initially hesitate to rely on AI-generated insights for strategic decisions.	Low	Medium	Provide training and position the system as a decision-support tool.
Scope Creep Risk	Stakeholders may request expanded monitoring capabilities during the MVP phase.	Medium	Medium	Maintain clear MVP scope and establish a roadmap for future features.

Risk Category	Risk Description	Probability	Impact	Mitigation Strategy
System Learning Curve Risk	Early outputs may require refinement as the system adapts to Lusitex's strategic priorities.	High	Medium	Implement feedback mechanisms and iterative improvements during early deployment.

8. Cost Analysis

The cost model evaluates implementation, infrastructure, and operational effort to estimate the three-year total cost of ownership.

This section estimates the cost of implementing and operating the Strategic Radar system for Lusitex Technical Fabrics. Two infrastructure approaches are evaluated:

- Open-source modular infrastructure stack
- Microsoft Azure vendor-managed infrastructure

The analysis includes implementation costs, operational infrastructure costs, human operational effort, and the estimated three-year Total Cost of Ownership (TCO). The goal is to provide a realistic view of the financial implications associated with deploying and maintaining the system.

All costs are expressed in euros (€).

8.1 Usage Assumptions

The cost model is based on realistic usage assumptions for Lusitex's intelligence needs during its European expansion.

Parameter	Value
Competitors monitored	7
Business sectors monitored	3

Parameter	Value
Technology monitoring	5 technologies
Initial intelligence run	One scan covering the previous 12 months
Monitoring frequency	Weekly
Users	5–10 internal users

Estimated intelligence output:

Period	Intelligence Briefs
Weekly	15
Monthly	~60
Yearly	~720

8.2 Implementation Costs

The Strategic Radar system requires an initial implementation phase estimated at approximately eight weeks. This phase includes the design and configuration of the intelligence system as well as preparation of the initial knowledge base.

Implementation activities include:

- strategic intelligence design
- competitor and sector identification
- knowledge base creation
- infrastructure configuration
- system testing and calibration
- training of Lusitex teams

8.2.1 Implementation Cost — Open Stack

Component	Cost (Open Stack)	Cost (Azure)
Strategic Radar consulting (8 weeks)	€14,500	€14,500
Azure cloud architecture consultant	NA	€5,500
Training and onboarding	€2,000	€2,000
Total Implementation Cost	**€16,500**	**€22,000**

8.3 Infrastructure Costs

Operational infrastructure costs include the services required to collect signals, run AI analysis, host the application, and deliver intelligence reports.

8.3.1 Open Stack Infrastructure

Cost Category	Component	Monthly Cost
Data	Tavily API	€30
Data	NewsAPI	€80
Model	OpenAI API usage	€80
Application	Pinecone vector database	€25
Operations	Railway hosting	€20
Operations	n8n automation	€25
Compliance	Monitoring / governance tools	€20
TOTAL	Total Infrastructure Cost	€280

8.3.2 Azure Infrastructure

Cost Category	Component	Monthly Cost
Data	Tavily API	€30
Data	NewsAPI	€80
Model	Azure OpenAI	€95
Application	Azure AI Search	€65
Operations	Azure App Service	€40
Operations	Azure Logic Apps	€30
Storage	Azure Blob Storage	€10
Compliance	Azure Monitor	€25
Total	Total Infrastructure Costs	€375

8.4 Operational Human Costs

After deployment, the Strategic Radar system requires ongoing operational supervision and periodic improvements.

Typical activities include:

- reviewing intelligence briefs
- validating signals and insights
- adjusting prompts and evaluation rules
- updating competitors and technologies monitored

Estimated operational effort:

Activity	Hours / Month	Cost
Operational monitoring	8 hours	€400
Continuous improvement	4 hours	€200
Total Human Operational Cost		€600 / month

Total Monthly Operational Cost:

Infrastructure	Infrastructure Cost	Human Cost	Total Monthly Cost
Open Stack	€280	€600	**€880**
Azure	€375	€600	**€975**

8.5 3-Year TCO

The system also performs a one-time intelligence scan of signals from the previous twelve months to build the initial knowledge baseline.

Initial Intelligence Scan Cost: €60

Infrastructure	Monthly Cost	Yearly Cost
Open Stack	€880	€10,560
Azure	€975	€11,700

<u>3-Year Total Cost of Ownership</u>

Infrastructure	Setup Cost	Initial Scan	3-Year Operations	3-Year TCO
Open Stack	€16,500	€60	€31,680	**€48,240**
Azure Infrastructure	€22,000	€60	€35,100	**€57,160**

8.6 Cost per Intelligence Brief

Based on an estimated output of approximately **60 intelligence briefs per month**, the cost per intelligence brief can be calculated as follows:

Open Stack

€880 / 60 ≈ **€15** per brief

Azure Infrastructure

€975 / 60 ≈ **€16** per brief

8.6.1 Interpretation

The open-stack architecture results in a lower three-year Total Cost of Ownership, mainly due to reduced infrastructure costs and the absence of specialized cloud architecture consulting.

The Azure platform provides advantages such as:

- integrated cloud ecosystem
- centralized monitoring and governance
- simplified infrastructure management

However, these operational advantages come with slightly higher long-term costs.

For Lusitex, the Strategic Radar system provides continuous strategic intelligence across European markets at a cost of approximately **€15–€16 per intelligence brief**, which is significantly lower than traditional market research or consulting analysis.

Even a single new contract, partnership, or procurement opportunity identified through the platform could generate value that exceeds the total cost of operating the system.

9. Implementation Roadmap

The deployment follows three phases: system deployment, operational monitoring, and continuous improvement.

The implementation of the Strategic Radar system is structured in three phases. This phased approach allows Lusitex to deploy the system quickly, establish an initial intelligence capability, and progressively refine the platform as the organization gains experience using AI-driven strategic intelligence.

The roadmap begins with the deployment of the system and the creation of an initial intelligence baseline. It then transitions into ongoing operational monitoring and continuous improvement of the intelligence processes.

9.1 Phase 1 — Deployment & Baseline Intelligence

The first phase focuses on deploying the Strategic Radar infrastructure and establishing the initial strategic intelligence environment. This phase is expected to take approximately **eight weeks** and results in a functional Minimum Viable Product (MVP).

Key activities in this phase include:

- Strategic intelligence design, including definition of competitors, monitored sectors, and technology watchlists
- Creation of the knowledge base containing company strategy, products, competitors, and technology priorities
- Infrastructure setup and configuration of the intelligence pipeline
- Integration of external data sources used for signal discovery
- Historical scan of the previous **12 months of market signals** to establish an initial intelligence baseline
- System testing, calibration of intelligence outputs, and training of Lusitex teams

At the end of this phase, Lusitex will have a fully operational Strategic Radar system capable of generating intelligence briefs.

9.2 Phase 2 — Operational Monitoring

Once the system is deployed, the project transitions into the operational monitoring phase. During this phase, the platform continuously collects and evaluates market signals relevant to Lusitex's strategic expansion in Europe.

Core activities include:

- Weekly monitoring of competitors, industry developments, and emerging textile technologies
- Detection of public procurement opportunities and relevant private-sector developments
- Generation of intelligence briefs summarizing key signals and potential strategic implications
- Periodic review and validation of the intelligence reports by Lusitex teams

This phase establishes Strategic Radar as an ongoing decision-support capability within the organization.

9.3 Phase 3 — Continuous Improvement

The final phase focuses on improving the system as Lusitex gains experience using the intelligence outputs. AI-based intelligence systems benefit from iterative refinement, particularly in how signals are evaluated and how intelligence briefs are generated.

Continuous improvement activities include:

- Refinement of prompts and evaluation criteria used in the intelligence analysis pipeline
- Adjustment of monitoring rules based on user feedback and business priorities
- Expansion of monitored sectors, competitors, and technology areas as the company enters new markets
- Integration of additional signal sources if new intelligence needs emerge

Over time, this phase allows the Strategic Radar platform to evolve into a more sophisticated intelligence capability aligned with Lusitex's international growth strategy.

10. Final Recommendation

Based on cost, flexibility, and organizational readiness, Lusitex should deploy Strategic Radar using the open-source infrastructure stack.

Lusitex Technical Fabrics aims to expand its presence in European markets and requires improved visibility into competitor activity, emerging textile technologies, and potential business opportunities. The Strategic Radar platform is designed to support this objective by continuously monitoring relevant signals and generating structured intelligence briefs for decision-makers.

Two infrastructure approaches were evaluated for deploying the system:

- Open-source modular infrastructure stack
- Microsoft Azure vendor-managed infrastructure

Both architectures are technically capable of supporting the Strategic Radar system. However, they differ in terms of cost structure, operational complexity, infrastructure flexibility, and degree of vendor dependency.

After evaluating the technical, organizational, and economic factors, the recommended approach for Lusitex is to **deploy the system initially using the Open Stack architecture.**

10.1 Comparison of the Two Approaches

Dimension	Open Stack	Azure Infrastructure
Implementation cost	Lower	Higher
Operational cost	Lower	Higher

Dimension	Open Stack	Azure Infrastructure
Infrastructure flexibility	High	Moderate
Vendor dependency	Low	High
Infrastructure management	Distributed services	Integrated platform
Compliance and monitoring	Limited built-in tools	Strong enterprise tools

The cost analysis shows that the Open Stack architecture results in a lower three-year Total Cost of Ownership.

Infrastructure	3-Year TCO
Open Stack	€48,240
Azure Infrastructure	€57,160

The Azure solution is therefore approximately 18% more expensive over a three-year period.

10.2 Reasons for the Recommended Approach

10.2.1 Cost Efficiency

As a mid-sized manufacturing company, Lusitex must balance innovation investments with financial sustainability. The Open Stack solution delivers the required functionality at a lower cost while still supporting the strategic objectives of the project.

10.2.2 Flexibility

The Strategic Radar system is expected to evolve as new competitors, technologies, sectors, and signal sources are added. A modular architecture allows Lusitex to adapt the platform over time without being constrained by a specific cloud ecosystem.

10.2.3 MVP-Oriented Deployment

The project is designed to deliver a **Minimum Viable Product (MVP)** within eight weeks. The Open Stack architecture enables rapid deployment using lightweight services and APIs, making it well suited for an iterative development approach.

10.2.4 Vendor Independence

By relying on modular services rather than a single cloud platform, Lusitex reduces the risk of vendor lock-in and maintains greater control over its technology stack.

10.3 Considerations and Trade-offs

Although the Open Stack architecture is recommended for the initial deployment, the Azure infrastructure approach offers several advantages that may become relevant as the system evolves.

These advantages include:

- integrated monitoring and governance tools
- enterprise-grade security frameworks
- simplified infrastructure management
- easier scaling for larger workloads

If the Strategic Radar platform grows significantly in scale or becomes integrated with other enterprise systems, Lusitex may consider migrating parts of the system to Azure or another enterprise cloud environment.

10.4 Strategic Impact

By implementing the Strategic Radar platform, Lusitex gains the ability to continuously monitor strategic developments across European markets.

The system enables the company to identify:

- new procurement opportunities
- emerging textile technologies
- competitor strategic moves
- relevant industry events and textile fairs

The cost analysis demonstrates that the platform can deliver actionable intelligence at a relatively low cost per insight. Even a single new contract, partnership, or procurement opportunity identified through the system could generate value that significantly exceeds the operational cost of the platform.

For these reasons, Strategic Radar represents a valuable capability for supporting Lusitex's international expansion strategy and strengthening the company's strategic awareness in competitive European markets.